

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	AIDS SLOT-B	Date:	15-08-2026

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. A company is developing an AI-powered customer support chatbot for banking services. The chatbot uses Context-Free Grammar (CFG) for parsing user queries. However, it faces issues such as:
Incorrect interpretation of ambiguous sentences

Failure to handle subject–verb agreement
Inefficient parsing for long queries Example
user query:

“Show me the transactions with the card from last month”

a) Analyze the ambiguity present in the given sentence using CFG

Ans:

The sentence is: “Show me the transactions with the card from last month.”

The phrase “with the card” can have more than one interpretation.

Interpretation 1: The customer wants transactions that were made using the card.

Interpretation 2: The customer wants transactions associated with a particular card.

The phrase “from last month” specifies the time period, but it may attach to the transactions or the card depending on the parse.

A simplified CFG representation is:

$S \rightarrow VP$

$VP \rightarrow V NP$

$NP \rightarrow Det N PP PP$

$PP \rightarrow P NP$

The ambiguity occurs because the prepositional phrases “with the card” and “from last month” can attach to different parts of the sentence

b) Evaluate the limitations of CFG in handling real-world conversational queries

Ans:

Basic CFG has several limitations:

1. It mainly represents syntactic structure and does not fully understand meaning.
2. Ambiguous sentences can produce multiple parse trees.
3. It does not naturally use conversational context.
4. Long sentences can produce many possible parses.
5. Basic CFG does not effectively handle subject–verb agreement.
6. It does not assign probabilities to different interpretations.
7. Informal or incomplete user queries may not match predefined grammar rules.

Therefore, CFG alone is not sufficient for a production banking chatbot

c) Evaluate and propose an improved solution using PCFG, feature structures, or Earley parsing

Ans:

A better solution is to combine CFG, Feature Structures, PCFG and Earley Parsing.

- **CFG** provides the basic syntactic grammar.
- **Feature Structures** handle grammatical features such as number and agreement.
- **PCFG** assigns probabilities to possible parse structures and selects the most likely interpretation.
- **Earley Parsing** efficiently handles ambiguous and long sentences.

For the given query, PCFG can assign a higher probability to the interpretation that matches common banking usage: transactions made using the card during the previous month.

d) Justify how your proposed method improves accuracy and efficiency in an industry setting.

Ans:

The proposed method improves the banking chatbot by:

- Reducing incorrect interpretations.
- Handling ambiguous queries better.
- Supporting long customer requests.
- Improving grammatical analysis.
- Selecting the most probable interpretation.

- Providing faster and more reliable responses.

Conclusion: Combining CFG with PCFG, Feature Structures and Earley Parsing provides better accuracy and efficiency than using basic CFG alone.

2. A voice assistant system (like Alexa/Siri-type applications) processes spoken commands using parsing techniques. The system currently uses top-down parsing, but faces:

Backtracking issues

Poor handling of incomplete or ambiguous input

Difficulty in real-time response Example

command:

“Book a flight to Delhi with a window seat”

a) Analyze how the given sentence can lead to multiple parse structures

Ans:

The command is: “Book a flight to Delhi with a window seat.”

The phrase “with a window seat” can be interpreted as a property of the flight booking or as an additional booking requirement.

One interpretation is: Book [a flight to Delhi] [with a window seat].

Here, the user wants a flight to Delhi and specifically wants a window seat.

Another possible interpretation is that the phrase “with a window seat” is attached directly to the flight object.

The ambiguity occurs because the prepositional phrase “with a window seat” can have different syntactic attachments.

b) Analyze the limitations of top-down parsing in this real-time application

Ans:

Top-down parsing starts from the start symbol and attempts to generate the input.

Its limitations include:

- It may require extensive backtracking.
- Ambiguous sentences create multiple possible paths.
- It can waste processing time on incorrect paths.
- Long spoken commands increase parsing complexity.
- Incomplete speech input can cause parsing failure.
- Real-time voice assistants require fast responses.

Therefore, basic top-down parsing may not be suitable for complex real-time voice commands.

c) Analyze how Earley parsing can improve handling of ambiguity and partial inputs

Ans:

Earley parsing is suitable for ambiguous and complex sentences because it can handle general CFG grammars efficiently.

For the given command, Earley parsing can maintain different possible interpretations while processing the sentence.

It can:

- Handle ambiguous structures.
- Process partial input.
- Reduce unnecessary backtracking.
- Support incomplete spoken commands.
- Work with complex CFG rules.
- Provide efficient parsing for real-time applications.

d) Compare and analyze the performance of top-down vs Earley parsing for such industry use cases

Ans:

Top-down parsing is simple but may suffer from repeated backtracking when multiple interpretations exist. Earley parsing maintains partial parsing states and can handle ambiguity more efficiently. For a voice assistant, Earley parsing is generally more suitable because spoken commands can be incomplete, ambiguous, or continuously received.

Conclusion: Earley parsing provides a more flexible and robust parsing approach for real-time voice assistant systems than basic top-down parsing.

3. A healthcare company is developing an AI system to analyze patient medical reports and automatically extract critical information such as diagnosis, prescribed treatments, and follow-up actions.

The system must handle:

Complex and lengthy medical sentences

Ambiguity in sentence structure

Accurate subject-verb agreement

Relationships between medical actions and entities (sub-categorization)

Real-time processing for hospital use Example input:

"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."

The current system, based only on basic Context-Free Grammar (CFG), fails to correctly interpret relationships and produces inconsistent outputs.

a) Design a complete NLP architecture integrating:

CFG for syntactic structure

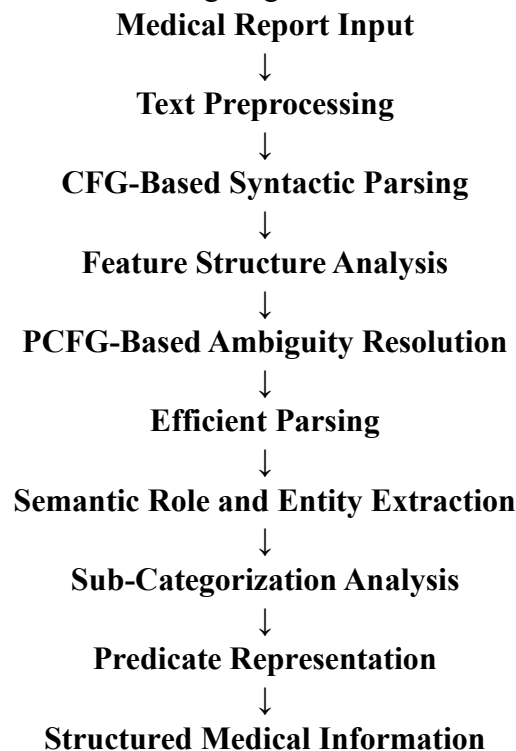
Probabilistic CFG (PCFG) for ambiguity resolution

Feature Structures for agreement handling

Efficient parsing techniques

Ans:

The proposed architecture contains the following stages:



The architecture combines syntactic, probabilistic and semantic processing.

CFG: CFG identifies the grammatical structure of the medical sentence.

PCFG: PCFG assigns probabilities to possible parse structures and selects the most likely interpretation.

Feature Structures

Feature structures represent grammatical information such as:

- Number
- Person
- Tense
- Agreement

This helps maintain correct subject–verb relationships.

Efficient Parsing: An efficient parser such as Earley parsing can be used to process complex and ambiguous medical sentences.

Semantic Processing

The system identifies:

- Medical entities.
- Actions.
- Relationships.
- Diagnosis.
- Treatment.

b) Develop a step-by-step workflow showing how the system processes the given medical sentence from input to structured output (diagnosis, action, location).

Ans:

Input sentence: “The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai.”

Step 1 – Input

The medical sentence is received by the NLP system.

Step 2 – Preprocessing

The sentence is tokenized and normalized.

Important terms include: doctor, reviewed, patient, recommends, medication, scheduling, follow-up visit, Chennai

Step 3 – CFG Parsing

CFG identifies the syntactic structure.

The main clause is: Doctor recommends medication and scheduling a follow-up visit.

The relative clause is: who reviewed the patient last week

Step 4 – Feature Structure Analysis

The system checks grammatical relationships.

Doctor → singular subject

recommends → singular verb

Therefore, subject–verb agreement is valid.

Step 5 – PCFG Processing

If multiple parse structures are possible, PCFG assigns probabilities and selects the most likely interpretation.

Step 6 – Semantic Extraction

The system identifies:

Doctor → Medical Entity

Patient → Medical Entity

Medication → Treatment

Follow-up visit → Medical Action

Chennai → Location

Step 7 – Predicate Representation

The relationships can be represented as:

Reviewed (Doctor, Patient, Last Week)

Recommends (Doctor, Medication)

Schedules (Follow Up Visit, Chennai)

Step 8 – Structured Output

The system extracts:

Diagnosis: Not explicitly provided.

Action: Start medication and schedule follow-up visit.

Location: Chennai.

This distinction is important because the sentence does not explicitly state a diagnosis. The system should not invent one.

c) Create a method to:

Incorporate sub-categorization frames for medical verbs

Handle real-time data processing

Improve accuracy and scalability in a hospital environment

Ans:

Medical verbs should have predefined semantic patterns.

For example:

recommend → Doctor + Treatment/Action

review → Doctor + Patient

schedule → Medical Staff + Appointment/Visit + Location

These frames help the system understand which entities can participate in a medical action.

Real-Time Processing

Real-time processing can be improved by:

- Using efficient parsing algorithms.
- Processing sentences incrementally.
- Using optimized medical NLP models.
- Caching frequently used medical terminology.
- Prioritizing critical medical information.
- Using parallel processing where appropriate.

Accuracy Improvement

Accuracy can be improved through:

- Medical-specific grammars.
- PCFG-based ambiguity resolution.
- Feature structures.
- Medical Named Entity Recognition.
- Semantic Role Labelling.
- Medical ontologies.
- Context-aware language models.
- Validation of extracted information.

Scalability in Hospitals

The system can scale by:

- Using modular NLP components.
- Processing reports in parallel.
- Using distributed computing.
- Maintaining medical terminology databases.
- Monitoring parsing and extraction accuracy.

- Integrating with hospital information systems.

Conclusion: A combined CFG, PCFG, Feature Structure and efficient parsing architecture can provide accurate and scalable medical information extraction. Sub-categorization frames further improve the understanding of relationships between medical actions and entities.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1								
2								
3								

Faculty In-charge	Course Coordinator	Program Director
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Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____